

Minecore specification

Kasparex codebase — derived from game config and hooks

Section

Column A	Column B	Column C	Column D	Column E	Column F
----------	----------	----------	----------	----------	----------

Document — Minecore specification (Kasparex codebase)
Generated (UTC) — 2026-04-26
Source paths — src/lib/game/minecore/*; hooks/useMinecore.ts; src/lib/games/registry.ts

Overview

Field	Value	Notes
Name / slug	Minecore / minecore	
Route	/games/minecore	
Registry version	0.1.0 (beta)	
Persistence key	minecore-state:{kaspaAddress guest}	localStorage
State version (initial)	1	
Mining reconnect grace	24 hours	RECONNECT_GRACE_MS
Refine rate	1 diamond !' 1 refinement point	MINECORE_REFINE_RATE
GRID redeemable field	Placeholder	gridRedeemableTotal not updated by apply-event
Default plant slots	4	MINECORE_DEFAULT_PLANT_SLOTS
Slot unlock cost	1 KAS	minecore:slot:unlock
Slot expand cost	50 KAS	minecore:slot:expand
Plant recharge	2.5 KAS per reserve unit	MINECORE_PLANT_RECHARGE_COST_KAS
Repair chance on extract	2%	Math.random() < 0.02
Expected diamonds formula	floor(baseOutput × worker × boost × batteryEff	computePlantExpectedDiamonds

Currencies

Resource	Role
Ingredients	8 types in IngredientBag (craft / power install)
Owned items	Counts: machines batteries workers modules
Diamonds	Balance + accumulated + live cycle diamonds
Refinement points	From Refine (1:1 in v1)
Shop	KAS (KREX tier discount) KREX GRID on some rows

PlantPresets

Type	Label	costKas	Description
standard	Standard Plant	0	Basic mining operations.
premium	Premium Plant	50	Upgraded infrastructure.
advanced	Advanced Complex	250	Industrial-scale mining.

Machines

id	label	durationMs	durationHuman	baseOutput	powerConsumptionFactor
pulse-drill	Pulse Drill	600000	10 min	50	1

id	label	durationMs	durationHuman	baseOutput	powerConsumptionFactor
crystal-extractor	Crystal Extractor	1800000	30 min	180	1.5
deep-vein-rig	Deep Vein Rig	3600000	60 min	420	2.5
quantum-fracturer	Quantum Fracturer	21600000	6 h	3200	6
NOTE	quantum-fracturer has no recipe in MINECORE_RECIPES				

Batteries

id	label	efficiency	powerCapacity	chargeCapacityMs	chargeHuman
energy-cell	Energy Cell	1	1	1800000	30 min
battery-pack	Battery Pack	1.15	2	3600000	60 min
diamond-capacitor	Diamond Capacitor	1.3	3	7200000	120 min
grid-battery	Grid Battery	1.5	4	21600000	360 min
NOTE	Only energy-cell is craftable in MINECORE_RECIPES				

PowerSources

id	label	kind	maxPowerUnits	drainRateMult	energyBudgetMult / installRequires
vein-thermal	Deep Vein Thermal	thermal	4	1.05	1.0 crystalDust 40 energyCells 2 circuitMesh 2
fission-bdag	Packed DAG Fission	fission	6	0.92	1.08 coreShards 2 alloyPlates 12 nullFragments 1 energyCells 3
krex-catalyst	Krex Catalytic Stack	catalytic	5	0.88	1.12 coolingGel 8 ariaChips 3 circuitMesh 4
aria-photon	ARIA Photonic Array	renewable	5	0.85	1.15 circuitMesh 6 energyCells 2 crystalDust 30
null-reactor	Null-Shell Reactor	exotic	3	0.8	1.2 nullFragments 2 coreShards 1 circuitMesh 8 alloyPlates 6

Modules

id	label	outputBonus	failureReduction
cooling-module	Cooling Module	0.05	0.05
stability-module	Stability Module	0.04	0.08
aria-sensor	ARIA Sensor	0.06	0.06
vector-drill-chip	Vector Drill Chip	0.08	0.04
NOTE	failureReduction not used in compute.ts yield		

Workers

id	label	multiplier
worker	Worker	1.1
operator	Operator	1.5
NOTE	Starter 1 worker 0 operator; no operator craft recipe	

Boosts

id	label	multiplier
none	No boost	1
krex-boost	KREX Boost	1.5
kas-overclock	KAS Overclock	2
grid-efficiency	GRID Efficiency	1.2

Recipes

recipeId	kind	outputId	requires
pulse-drill	machine	pulse-drill	alloyPlates 10 circuitMesh 5 energyCells 2
crystal-extractor	machine	crystal-extractor	alloyPlates 18 circuitMesh 10 ariaChips 2 energyCells 4
deep-vein-rig	machine	deep-vein-rig	alloyPlates 35 circuitMesh 16 coreShards 3 energyCells 6
energy-cell	battery	energy-cell	circuitMesh 2 energyCells 1
cooling-module	module	cooling-module	coolingGel 6 alloyPlates 4
stability-module	module	stability-module	alloyPlates 6 circuitMesh 4
aria-sensor	module	aria-sensor	ariaChips 4 circuitMesh 4
vector-drill-chip	module	vector-drill-chip	circuitMesh 8 nullFragments 1

StarterIngredients

ingredient	amount
crystalDust	200
alloyPlates	60
circuitMesh	40
energyCells	30
coreShards	10
coolingGel	25
ariaChips	15
nullFragments	5

MiningCycle

Rule	Behavior
StartMining	Costs 1 reserve unit; needs setup power charge > 0 no active cycle
Live diamonds	Linear in time vs durationMs; capped by battery runtime and cycle end
StopMining	Pause: no drain; Resume extends endAtMs
ExtractionReady	now >= endAtMs if battery not empty earlier
BatteryEmpty	Charge 0 before cycle end
Extract	Banks accumulated + live; clears cycle; 2% needsRepair
Refine mid-cycle	mintedOffset; full siphon before natural end refunds 1 reserve unit

ShopUI

itemId	title	baseKas	Other prices / notes
crystalDust	Crystal Dust	0.5	KREX 0.5
alloyPlates	Alloy Plates	2	KREX 1.2
circuitMesh	Circuit Mesh	1.5	KREX 1.0 GRID 0.4
energyCells	Energy Cells	3	KREX 2.5 GRID 0.3
kas-overclock	KAS Overclock	5	KAS only
krex-boost	KREX Boost	0	KREX 25 UI disabled Soon
power-topup	Plant recharge	2.5	Per unit + full battery
repair	Stability Patch	2	KAS

PaidKAS

Action	Amount / pattern	skuld / notes
Plant recharge	2.5 KAS × units	minecore:plant:recharge
Slot unlock	1 KAS	minecore:slot:unlock
Slot expand	50 KAS	minecore:slot:expand
Plant upgrade	50 / 250 KAS	minecore:upgrade:{slot}:{type}
Ingredient buy	line total	minecore:ingredient:{name}
Power top-up		minecore:power:topup
Battery refill		minecore:battery:refill
Repair		minecore:repair
KAS discount		KREX tier discount in useMinecore

NFT Automation

Topic	Detail
Slot types	worker operator foreman engineer
foremanActive	Any foreman slot has NFT
autoRestart	Extract ExtractionReady; StartMining ReadyToMine
Foreman refill	Intended energy cell + RefillBattery; AddIngredients clamps negatives

PlantStatus

status	Meaning
EmptySlot	Not unlocked
SetupIncomplete	Missing machine battery or worker
ReadyToMine	Setup OK reserve > 0 no cycle
MiningActive	Cycle running not paused battery > 0
MiningPaused	Paused
BatteryEmpty	Mid-cycle battery depleted
ExtractionReady	Cycle window ended
NeedsRepair	Repair flag set
NeedsPower	powerRemaining <= 0 no cycle

Redeem Ledger

Topic	Detail
Refine	1:1 to points; grid ledger last 200 rows
RedeemGrid event	Subtracts refinement points only
API	GET /api/games/minecore/grid-ledger?address=